

All-Angle Harmonic Strobes: Classical Resonance, Gaussian Koopman Spectrum, and Same-Clock Quantization

Route-A structural certificate C178

Abstract

For the planar oscillator $H = (q^2 + p^2)/2$, we give one exact spectral ledger for every physical time $\theta \in \mathbb{R}$. The n th iterate is a rotation through $n\theta$: it fixes only the origin off resonance and the whole plane at resonance. Consequently, irrational $\theta/(2\pi)$ has Artin–Mazur zeta $(1 - z)^{-1}$, whereas each rational angle has infinite fixed sets and no ordinary Artin–Mazur series. On invariant Gaussian L^2 , a Laguerre–angular orthonormal basis diagonalizes the Koopman unitary with eigenvalues $e^{im\theta}$ and infinite radial multiplicity. The natural oscillator quantization at the same clock is diagonal in the Hermite basis and is a 4π -periodic metaplectic lift: a 2π shift produces the global sign -1 . It satisfies exact Egorov and reversal but, like the Koopman unitary, is noncompact, non-Schatten, and has no ordinary trace-class Fredholm determinant. Thus natural quantization is genuine progress yet cannot repair the failed periodic-orbit and determinant gates.

Keywords: harmonic oscillator; stroboscopic rotation; Gaussian Koopman spectrum; Laguerre basis; exact Egorov; Route A.

中文摘要

对平面谐振子 $H = (q^2 + p^2)/2$ ，本文在所有物理实时间 $\theta \in \mathbb{R}$ 上建立统一的精确谱账本。第 n 次迭代是转角 $n\theta$ 的旋转：非共振时仅固定原点，共振时固定整个平面。因此，无理角的 Artin–Mazur 动力 ζ 函数为 $(1 - z)^{-1}$ ，而每个有理角均出现无限不动点集，普通定义失效。在不变高斯平方可积空间中，Laguerre–角向正交基把 Koopman 酉算子完全对角化，其特征值为 $e^{im\theta}$ 且具有无限径向重数。同一物理时钟下的自然谐振子量子化由 Hermite 基对角化。该量子提升具有 4π 周期，时间平移 2π 会保留不可忽略的全局负号；同时它满足精确 Egorov 关系和共轭反演。但两个酉算子都非紧、非有限 Schatten 类，故无普通迹类 Fredholm 行列式。自然量子化不能弥补周期轨道与行列式闸门的失败。

关键词：谐振子；频闪旋转；高斯 Koopman 谱；Laguerre 基；精确 Egorov 关系；路线 A。

1 All-angle classical theorem

Freeze physical time as $\theta \in \mathbb{R}$. Hamilton’s equations give

$$T_\theta(q, p) = (q \cos \theta + p \sin \theta, -q \sin \theta + p \cos \theta), \quad T_\theta^n = T_{n\theta}, \quad T_{\theta+2\pi} = T_\theta.$$

Hence, for every $n \geq 1$,

$$\text{Fix}(T_\theta^n) = \begin{cases} \mathbb{R}^2, & n\theta \in 2\pi\mathbb{Z}, \\ \{(0, 0)\}, & n\theta \notin 2\pi\mathbb{Z}. \end{cases} \quad (1)$$

Indeed, a nonidentity planar rotation fixes only the origin, whereas the identity fixes the plane. If $\theta/(2\pi)$ is irrational, all fixed counts equal one, and the formal identity

$$\zeta_{\text{AM}}(z) = \exp\left(\sum_{n \geq 1} \frac{z^n}{n}\right) = (1 - z)^{-1} \quad (2)$$

follows from $-\log(1 - z) = \sum_{n \geq 1} z^n/n$. If $\theta/(2\pi) = a/b$ in lowest terms, (1) equals $\mathbb{R}^2$ exactly when $b \mid n$; these uncountable cardinalities are not finite Artin–Mazur coefficients. Thus the ordinary series is undefined, including $b = 1$. For $b > 1$, every nonzero point has least period b . Finally, $S(q, p) = (q, -p)$ is an involution with $ST_\theta S = T_\theta^{-1}$.

2 Exact Gaussian Koopman spectrum

Write $q - ip = re^{i\varphi}$, so T_θ advances φ by θ , and freeze

$$d\gamma = \pi^{-1} e^{-r^2} dq dp, \quad U_\theta f = f \circ T_\theta.$$

Rotation preserves γ , hence U_θ is unitary and $U_{\theta+2\pi} = U_\theta$. For $m \in \mathbb{Z}$ and $k \geq 0$, define

$$\psi_{k,m} = \sqrt{\frac{k!}{(k + |m|)!}} r^{|m|} L_k^{|m|}(r^2) e^{im\varphi}.$$

The angular Fourier basis and, after $x = r^2$, the identity

$$\int_0^\infty e^{-x} x^{|m|} L_k^{|m|}(x) L_\ell^{|m|}(x) dx = \frac{(k + |m|)!}{k!} \delta_{k\ell}$$

prove orthonormality and completeness. Direct substitution gives the all-parameter diagonalization

$$U_\theta \psi_{k,m} = e^{im\theta} \psi_{k,m}. \quad (3)$$

For irrational $\theta/(2\pi)$, the eigenvalues indexed by m are distinct and dense on the unit circle. For reduced a/b , they are exactly the b th roots of unity. In both cases, varying k gives each eigenvalue countably infinite multiplicity. Thus U_θ is noncompact; all its singular values are one, so it lies in no finite Schatten class and $\det(I - zU_\theta)$ is not an ordinary trace-class Fredholm determinant for $z \neq 0$. If $V_S f = f \circ S$ and K denotes complex conjugation, then the antiunitary $\Theta_G = V_S K$ obeys $\Theta_G U_\theta \Theta_G^{-1} = U_\theta^{-1}$.

3 Natural same-clock quantum lift

On $L^2(\mathbb{R})$, let

$$\hat{H} = \frac{1}{2} \left(-\frac{d^2}{dx^2} + x^2 \right), \quad Q_\theta = e^{-i\theta \hat{H}}.$$

The standard Hermite basis gives

$$\hat{H} h_j = (j + \frac{1}{2}) h_j, \quad Q_\theta h_j = e^{-i\theta(j+1/2)} h_j. \quad (4)$$

Here θ remains real physical time; it is not replaced by a class modulo 2π . Equation (4) gives

$$Q_{\theta+2\pi} = -Q_\theta, \quad Q_{\theta+4\pi} = Q_\theta. \quad (5)$$

Thus the metaplectic lift is 4π -periodic and only projectively 2π -periodic; the global sign is retained. Irrational times give distinct dense pure point phases. For the exact real-time representative $\theta/(2\pi) = a/b$ in lowest terms, the spectrum is $e^{-i\pi a/b}$ times the b th roots, each with infinite multiplicity. Replacing a by $a + b$ multiplies Q_θ by -1 . On the Schwartz core, with $\hat{p} = -i d/dx$,

$$[\hat{H}, \hat{q}] = -i\hat{p}, \quad [\hat{H}, \hat{p}] = i\hat{q}.$$

Integrating the Heisenberg equations proves the exact, same-clock Egorov law

$$Q_\theta^* \hat{q} Q_\theta = \hat{q} \cos \theta + \hat{p} \sin \theta, \quad Q_\theta^* \hat{p} Q_\theta = -\hat{q} \sin \theta + \hat{p} \cos \theta. \quad (6)$$

Complex conjugation satisfies $K Q_\theta K = Q_\theta^{-1}$. Equation (4) also supplies an orthonormal image sequence, so Q_θ is noncompact, belongs to no finite Schatten class, and has no ordinary trace-class Fredholm determinant. Although $e^{-t\hat{H}}$ is trace class for $t > 0$, it uses imaginary time. Heat damping, Wick rotation, and Hermite truncation are different clocks or objects and cannot serve as determinants of T_θ , U_θ , or Q_θ .

4 Route-A decision and evidence boundary

Gate	Verdict	Exact reason
A0	FAIL	no intrinsic arithmetic origin or finite all-angle orbit carrier
A1	FAIL	trivial irrational ledger; rational fixed sets are infinite
A2	FAIL	ordinary Koopman and quantum determinants are unavailable
A3	FAIL	no target divisor, functional equation, or counting-law match
A4	NATURAL_QUANTIZATION	real-time metaplectic lift, Egorov, and reversal

The exact tuple is

$$(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION).$$

The overall verdict is `ROUTE_A_REJECTED`, and Route B is false. The positive A4 theorem cannot compensate for A0–A3.

Finite regression ledgers contain 1,656 rational fixed-set rows, 108 irrational rows, 209 Laguerre rows, 874 Koopman-phase rows, and 736 quantum phase rows. A producer-independent checker passes 26,271 assertions, SymPy passes 10,465 exact checks, byte replay is exact, and all 65 hostile mutations are rejected. These sentinels test implementations; the all-parameter results rest on the proofs above. No external novelty, priority, reviewer, or acceptance claim is made.

Limitations. We use ordinary Artin–Mazur and trace-class Fredholm definitions, do not introduce regularized determinants, and never identify $Q_{\theta+2\pi}$ with Q_θ as unitaries. No target zero or prime table, fitted parameter, arithmetic local datum, Euler factor, root number, automorphy object, Hilbert–Pólya operator, or Route-B authorization enters the package. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript. **Ethics.** No human, animal, clinical, personal, private, or sensitive data are used; approval is not applicable. **Author contributions (CRediT).** This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing; AI systems are not authors. **Funding.** No external funding is reported. **Conflicts of interest.** None known. **AI-use disclosure.** An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or an independent error process.